Nmap one of the most popular network mappers in the infosec world. It's utilized by cybersecurity professionals and newbies alike to audit and discover local and remote open ports, as well as hosts and network information.

Some of this tool's best features are that it's open-source, free, multi-platform and receives constant updates each year. It also has a big plus: it's one of the most complete host and network scanners available. It includes a large set of options to enhance your scanning and mapping tasks, and brings with it an incredible community and comprehensive documentation to help you understand this tool from the very start. Nmap can be used to:

- Create a complete computer network map.
- Find remote IP addresses of any hosts.
- Get the OS system and software details.
- Detect open ports on local and remote systems.
- Audit server security standards.
- Find vulnerabilities on remote and local hosts.

It was mentioned in the Top 20 OSINT Tools article we published, and today we'll explore a little bit more about this essential security tool with some practical terminal-based Nmap commands.

Best 15 Nmap command examples

Let's get to know a few useful command-line based scans that can be performed using Nmap.

1. Basic Nmap Scan against IP or host

```
nmap 1.1.1.1
```

Now, if you want to scan a hostname, simply replace the IP for the host, as you see below:

```
nmap cloudflare.com
```

These kinds of basic scans are perfect for your first steps when starting with Nmap.

2. Scan specific ports or scan entire port ranges on a local or remote server

```
nmap -p 1-65535 localhost
```

In this example, we scanned all 65535 ports for our localhost computer.

Nmap is able to scan all possible ports, but you can also scan specific ports, which will report faster results. See below:

```
nmap -p 80,443 8.8.8.8
```

3. Scan multiple IP addresses

Let's try to scan multiple IP addresses. For this you need to use this syntax:

```
nmap 1.1.1.1 8.8.8.8
```

You can also scan consecutive IP addresses:

```
nmap -p 1.1.1.1,2,3,4
```

This will scan `1.1.1.1`, `1.1.1.2`, `1.1.1.3` and `1.1.1.4`.

4. Scan IP ranges

You can also use Nmap to scan entire CIDR IP ranges, for example:

```
nmap -p 8.8.8.0/28
```

This will scan 14 consecutive IP ranges, from `8.8.8.1` to `8.8.8.14`.

An alternative is to simply use this kind of range:

```
nmap 8.8.8.1-14
```

You can even use wildcards to scan the entire C class IP range, for example:

```
nmap 8.8.8.*
```

This will scan 256 IP addresses from `8.8.8.1` to `8.8.8.256`.

If you ever need to exclude certain IPs from the IP range scan, you can use the "–exclude" option, as you see below:

```
nmap -p 8.8.8.* --exclude 8.8.8.1
```

5. Scan the most popular ports

Using "–top-ports" parameter along with a specific number lets you scan the top X most common ports for that host, as we can see:

```
nmap --top-ports 20 192.168.1.106
```

Replace "20" with the desired number. Output example:

```
[root@securitytrails:~]nmap --top-ports 20 localhost
Starting Nmap 6.40 ( http://nmap.org ) at 2018-10-01 10:02 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000016s latency).
```

```
Other addresses for localhost (not scanned): 127.0.0.1
PORT     STATE    SERVICE
21/tcp   closed   ftp
22/tcp   closed   ssh
23/tcp   closed   telnet
25/tcp   closed   smtp
53/tcp   closed   domain
80/tcp   filtered http
110/tcp  closed   pop3
111/tcp  closed   rpcbind
135/tcp  closed   msrpc
139/tcp  closed   netbios-ssn
143/tcp  closed   imap
443/tcp  filtered https
445/tcp  closed   microsoft-ds
993/tcp  closed   imaps
995/tcp  closed   pop3s
1723/tcp closed   pptp
3306/tcp closed   mysql
3389/tcp closed   ms-wbt-server
5900/tcp closed   vnc
8080/tcp closed   http-proxy
```

6. Scan hosts and IP addresses reading from a text file

In this case, Nmap is also useful to read files that contain hosts and IPs inside.

Let's suppose you create a list.txt file that contains these lines inside:

```
192.168.1.106
cloudflare.com
microsoft.com
securitytrails.com
```

The "-iL" parameter lets you read from that file, and scan all those hosts for you:

```
nmap -iL list.txt
```

7. Save your Nmap scan results to a file

On the other hand, in the following example we will not be reading from a file, but exporting/saving our results into a text file:

```
nmap -oN output.txt securitytrails.com
```

Nmap has the ability to export files into XML format as well, see the next example:

```
nmap -oX output.xml securitytrails.com
```

8. Disabling DNS name resolution

If you need to speed up your scans a little bit, you can always choose to disable reverse DNS resolution for all your scans. Just add the “-n” parameter.

```
[root@securitytrails:~]nmap -p 80 -n 8.8.8.8
Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 09:15 -03
Nmap scan report for 8.8.8.8
Host is up (0.014s latency).
PORT   STATE    SERVICE
80/tcp filtered http
```

See the difference with a normal DNS-resolution enabled scan:

```
[root@securitytrails:~]nmap -p 80 8.8.8.8
Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 09:15 -03
Nmap scan report for google-public-dns-a.google.com (8.8.8.8)
Host is up (0.014s latency).
PORT   STATE    SERVICE
80/tcp filtered http
```

9. Scan + OS and service detection with fast execution

Using the “-A” parameter enables you to perform OS and service detection, and at the same time we are combining this with “-T4” for faster execution. See the example below:

```
nmap -A -T4 cloudflare.com
```

This is the output we got for this test:

```
[root@securitytrails:~]nmap -A -T4 cloudflare.com

Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 08:57 -03
Nmap scan report for cloudflare.com (198.41.214.162)
Host is up (0.022s latency).
Other addresses for cloudflare.com (not scanned): 198.41.215.162 2400:cb00:2048:1::c629:d7a2 2400:cb00:2048:1::c629:d6a2
Not shown: 996 filtered ports
PORT     STATE SERVICE       VERSION
80/tcp   open  http          Cloudflare nginx
| http-server-header:
|   cloudflare
|_  cloudflare-nginx
|_http-title: Did not follow redirect to https://www.cloudflare.com/
443/tcp  open  ssl/https     cloudflare
| fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.1 403 Forbidden
|     Server: cloudflare
|     Date: Mon, 01 Oct 2018 11:58:15 GMT
|     Content-Type: text/html
|     Content-Length: 167
|     Connection: close
|     CF-RAY: 462ec1a4696267c1-EZE
|     <html>
|     <head><title>403 Forbidden</title></head>
|     <body bgcolor="white">
|     <center><h1>403 Forbidden</h1></center>
|     <hr><center>cloudflare</center>
|     </body>
|     </html>
```

10. Detect service/daemon versions

This can be done by using -sV parameters

```
nmap -sV localhost
```

As you can see here:

```
[root@securitytrails:~]nmap -sV localhost
Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 09:28 -03
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000020s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed ports
PORT STATE SERVICE VERSION
111/tcp open rpcbind 2-4 (RPC #100000)
631/tcp open ipp CUPS 2.2
902/tcp open ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC,
SOAP)

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.96 seconds
```

11. Scan using TCP or UDP protocols

One of the things we love most about Nmap is the fact that it works for both TCP and UDP protocols. And while most services run on TCP, you can also get a great advantage by scanning UDP-based services. Let's see some examples.

Standard TCP scanning output:

```
[root@securitytrails:~]nmap -sT 192.168.1.1
Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 09:33 -03
Nmap scan report for 192.168.1.1
Host is up (0.58s latency).
Not shown: 995 closed ports
PORT STATE SERVICE
80/tcp open http
1900/tcp open upnp
20005/tcp open btx
49152/tcp open unknown
49153/tcp open unknown
Nmap done: 1 IP address (1 host up) scanned in 1.43 seconds
```

UDP scanning results using "-sU" parameter:

```
[root@securitytrails:~]nmap -sU localhost
Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 09:37 -03
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000021s latency).
Other addresses for localhost (not scanned): ::1
```

```
Not shown: 997 closed ports
PORT STATE SERVICE
68/udp open|filtered dhcpc
111/udp open rpcbind
5353/udp open|filtered zeroconf
```

12. CVE detection using Nmap

One of Nmap's greatest features that not all the network and systems administrators know about is something called "Nmap Scripting Engine" (known as NSE). This scripting engine allows users to use a pre-defined set of scripts, or write their own using Lua programming language.

Using NSE is crucial in order to automate system and vulnerability scans. For example, if you want to run a full vulnerability test against your target, you can use these parameters:

```
nmap -Pn --script vuln 192.168.1.105
```

Output example:

```
[root@securitytrails:~]nmap -Pn --script vuln 192.168.1.105
Starting Nmap 7.60 ( https://nmap.org ) at 2018-10-01 09:46 -03
Pre-scan script results:
| broadcast-avahi-dos:
| Discovered hosts:
| 224.0.0.251
| After NULL UDP avahi packet DoS (CVE-2011-1002).
|_ Hosts are all up (not vulnerable).
Nmap scan report for 192.168.1.105
Host is up (0.00032s latency).
Not shown: 995 closed ports
PORT STATE SERVICE
80/tcp open http
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
| http-slowloris-check:
| VULNERABLE:
| Slowloris DOS attack
| State: LIKELY VULNERABLE
| IDs: CVE:CVE-2007-6750
| Slowloris tries to keep many connections to the target web server open and
hold
| them open as long as possible. It accomplishes this by opening connections
to
| the target web server and sending a partial request. By doing so, it
starves
| the http server's resources causing Denial Of Service.
|
| Disclosure date: 2009-09-17
| References:
| http://ha.ckers.org/slowloris/
|_ https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
1900/tcp open upnp
```

```
20005/tcp open btx
49152/tcp open unknown
49153/tcp open unknown
```

13. Launching DOS with Nmap

Nmap features never seem to end, and thanks to the NSE, that even allows us to launch DOS attacks against our network testings.

In our previous example (#12) we found the host was vulnerable to Slowloris attack, and now we'll try to exploit that vulnerability by launching a DOS attack in a forever loop:

```
nmap 192.168.1.105 -max-parallelism 800 -Pn --script http-slowloris --script-
args http-slowloris.runforever=true
```

14. Launching brute force attacks

NSE is really fascinating – it contains scripts for everything you can imagine. See the next three examples of BFA against WordPress, MSSQL, and FTP server:

WordPress brute force attack:

```
nmap -sV --script http-wordpress-brute --script-args
'userdb=users.txt,passdb=passwds.txt,http-wordpress-
brute.hostname=domain.com, http-wordpress-
brute.threads=3,brute.firstonly=true' 192.168.1.105
```

Brute force attack against MS-SQL:

```
nmap -p 1433 --script ms-sql-brute --script-args
userdb=customuser.txt,passdb=custompass.txt 192.168.1.105
```

FTP brute force attack:

```
nmap --script ftp-brute -p 21 192.168.1.105
```

15. Detecting malware infections on remote hosts

Nmap is able to detect malware and backdoors by running extensive tests on a few popular OS services like on Identd, Proftpd, Vsftpd, IRC, SMB, and SMTP. It also has a module to check for popular malware signs inside remote servers and integrates Google's Safe Browsing and VirusTotal databases as well.

A common malware scan can be performed by using:

```
nmap -sV --script=http-malware-host 192.168.1.105
```

Or using Google's Malware check:

```
nmap -p80 --script http-google-malware infectedsite.com
```

Output example:

```
80/tcp open  http
|_http-google-malware.nse: Host is known for distributing malware.
```

Nmap is one of the most complete and accurate port scanners used by infosec professionals today. With it, you can perform simple port scan tasks or use its powerful scripting engine to launch DOS attacks, detect malware or brute force testings on remote and local servers.

Today we covered the top fifteen Nmap commands to scan remote hosts, but there's a lot more to discover if you're starting to use Nmap in your OSINT strategy.

1. TCP Full Open Scan (TCP Connect Scan)

Completes the full TCP 3-way handshake

nmap -sT 192.168.1.10

How it works:
SYN → SYN-ACK → ACK
✓ Reliable
✕ Easily detected & logged

Use when: You don't have raw socket privileges.

2. TCP Half Open Scan (SYN Scan)

Also called **Stealth SYN scan**

nmap -sS 192.168.1.10

How it works:
SYN → SYN-ACK → RST (connection not completed)

✓ Fast
✓ Less logged than full scan
✓ Most commonly used scan

3. Stealth Scan (General Term)

Usually refers to **SYN scan**

nmap -sS 192.168.1.10

Why "stealth":
Avoids completing the TCP handshake, so fewer application logs.

4. XMAS Scan

Sets **FIN, PSH, and URG** flags (like a "Christmas tree" 🎄)

nmap -sX 192.168.1.10

Response behavior:

- Open port → No response
- Closed port → RST

✓ Can bypass some firewalls
✕ Doesn't work well on Windows targets

5. FIN Scan

Sends only the **FIN flag**

nmap -sF 192.168.1.10

Response behavior:

- Open → No response
- Closed → RST

✓ Useful for firewall evasion
✕ Unreliable on non-RFC-compliant systems

6. ACK Scan

Used to **map firewall rules**, not to find open ports

nmap -sA 192.168.1.10

What it tells you:

- RST → Port is **unfiltered**
- No response → Port is **filtered**

✓ Identifies firewall behavior
✕ Does not show open/closed ports

7. Idle Scan (Zombie Scan)

Highly stealthy scan using a **zombie host**

nmap -sI 192.168.1.20 192.168.1.10

- 192.168.1.20 → Zombie host
- 192.168.1.10 → Target

✓ Extremely stealthy
✓ Target never sees your IP
✕ Requires an idle zombie system

Quick Comparison Table

Scan Type	Command	Purpose
TCP Full Open	-sT	Reliable but noisy
Half Open / SYN	-sS	Fast, stealthy
Stealth Scan	-sS	Avoids full handshake
XMAS Scan	-sX	Firewall evasion
FIN Scan	-sF	Stealth probing
ACK Scan	-sA	Firewall mapping
Idle Scan	-sI	Anonymous scanning